

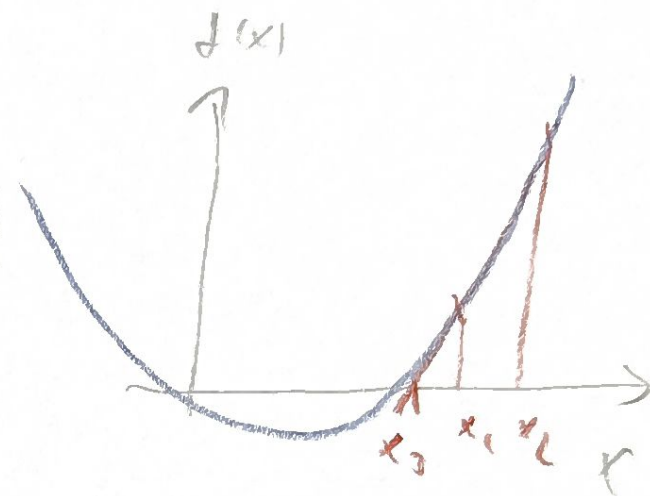
Introduction to numerical Methods - 9 -

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False position method (regula falsi)

Predict the position of the root by
linear approximation

$$\frac{f(x_3) - f(x_L)}{x_3 - x_L} \approx \frac{f(x_1) - f(x_L)}{x_1 - x_L}$$



Set $f(x_3) = 0 \Rightarrow$

$$x_3 = x_L - f(x_L) \frac{x_1 - x_L}{f(x_1) - f(x_L)}$$

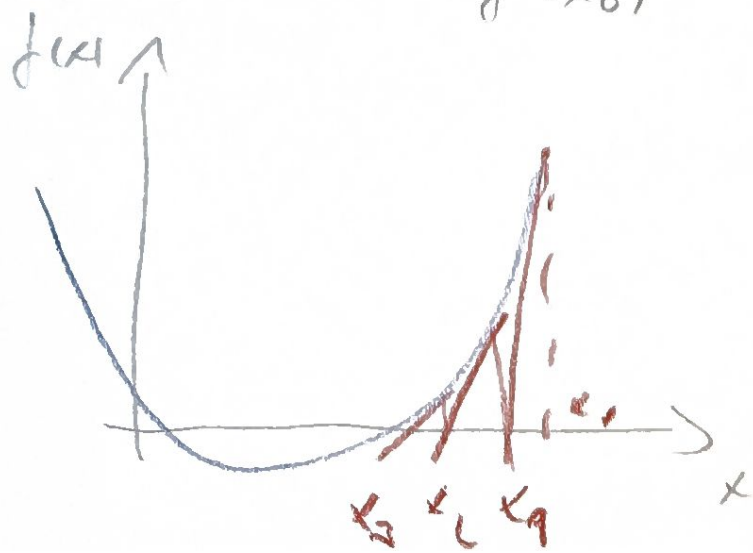
- Much faster than bisection
- Works with starting interval without a root

Newton's method: single point method,

use the local gradient

$$0 = f(x_0) + f'(x_0)(x_1 - x_0)$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$



- generally faster than
regula falsi
- requires derivative

— If you know how to find minima then finding minima of $f^2(x)$ can give you the roots

Finding the minima

Gradient method:

Let $f(x)$ be a differentiable, and $\delta > 0$

then

$$x_{n+1} = x_n - \delta f'(x_n)$$

if x_n is not an extremal point, then

$f(x_{n+1}) \leq f(x_n)$ if δ is sufficiently small

Conjugate gradient method

idea: close to the minima everything is a parabola: $f(x) \approx \beta(x-a)^2$
minimum a .

Two points x_1, x_2 , value $f(x_1), f(x_2)$
derivative $f'(x_1), f'(x_2)$

The gradient method:

$$x_3 = x_2 - \gamma f'(x_2) = x_2 - 2\gamma\beta(x_2 - a)$$

We want $x_3 = a \Rightarrow 2\gamma\beta = 1 \Rightarrow \gamma = \frac{1}{2\beta}$

So if we can make $\gamma = \frac{1}{2\beta}$ the gradient method will go to the minimum

$$f'(x_1) = 2\beta(x_1 - a) \quad f'(x_2) = 2\beta(x_2 - a) \quad | -$$

$$f'(x_2) - f'(x_1) = 2\beta x_2 - 2\beta x_1 = 2\beta(x_2 - x_1)$$

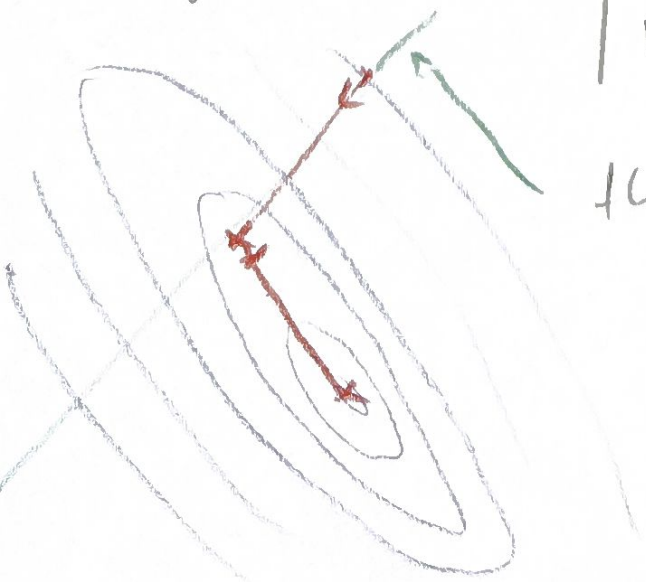
$$\gamma = \frac{1}{2\beta} = \frac{x_2 - x_1}{f'(x_2) - f'(x_1)}$$

Conjugate method in n dimensions

$$\underline{x}_{n+1} = \underline{x}_n - \gamma_n \nabla f(\underline{x}_n)$$

$$\gamma_n = \frac{(\underline{x}_n - \underline{x}_{n-1})^T (\nabla f(\underline{x}_n) - \nabla f(\underline{x}_{n-1}))}{\|\nabla f(\underline{x}_n) - \nabla f(\underline{x}_{n-1})\|^2}$$

this art is also a parabola



Problems

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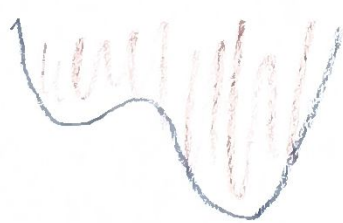
- local minima
- = stochastic processes
- etc.

Simulated annealing

- stochastic method
- able to break from local minimum

Idea: - allow unfavourable moves

- slowly cool down the system:
at large temperatures, the system does not care about the energy landscape, but as cooled gradually goes to valleys but has time to escape local minima



x_{new}



- ergodic moves: $\uparrow$ may reach all possible states

- x_{new} :

• $f(x_{new}) \leq f(x_{old})$: keep new pos

• $f(x_{new}) > f(x_{old})$: keep new pos with

prob. $\exp(- (f(x_{new}) - f(x_{old})) / T)$

T temp and goes to 0 for $t \rightarrow \infty$